BDA Assignments

Weekly Business Review – E-commerce Marketplace

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1. Introduction

This report is a Weekly Business Review of the Brazilian Olist e-commerce dataset for our Big Data Analytics assignment. The dataset contains about 100,000 real marketplace orders placed between late 2016 and mid 2018. Over this period, total sales reached **R$ 13.22 Million**, peaking at R$ 992.8k in May 2018.

Even though the marketplace grew steadily, the data reveals two major operational problems:
• **Almost no repeat customers:** Only **3.04%** of customers made a second purchase. Nearly 97% of buyers bought once and never returned.
• **Delivery delays hurt customer reviews:** While 92.20% of orders were delivered on time, **36.60% of 1-star reviews had late deliveries**, arriving an average of **12.36 days late**.

2. Data Quality Checks

Before doing the analysis, we tested the database tables using automated SQL data quality checks in PostgreSQL:

Check Name	Failed Rows	Status	Explanation
Duplicate Order IDs	**0**	**PASS**	Every order appears exactly once in the order summary mart.
Null Order IDs	**0**	**PASS**	No missing order IDs were found.
Delivered Before Purchase	**0**	**PASS**	Delivery timestamp is always after purchase timestamp.
Review Scores Outside 1–5	**0**	**PASS**	All customer review ratings are valid integers between 1 and 5.
Missing Customer Records	**0**	**PASS**	All orders link correctly to a valid customer record.
Payment Amount Mismatches	**300**	**WARN**	300 orders have small differences due to discounts, vouchers, or installment interest. Kept as warnings without changing data.

3. Main Metric and Supporting Metrics

Metric Type	Metric Name	Value	Why It Matters
Main Metric	**Total Sales (GMV)**	**R$ 13.22M**	Total value of all items sold, excluding freight fees and cancelled orders.
Supporting	**Repeat Purchase Rate**	**3.04%**	Measures how many customers buy again. Only 2,888 out of 94,990 customers ever returned.
Supporting	**On-Time Delivery Rate**	**92.20%**	Percentage of delivered orders that arrived on or before the promised delivery date.
Guardrail	**Average Days Late**	**+11.8 days**	Average delay when an order is late. Double-digit delays cause severe customer complaints.
Guardrail	**Average Review Score**	**4.09 / 5.0**	Average rating across all customer reviews. Helps track overall customer satisfaction.

4. Monthly Sales Trend

Using SQL query 04_monthly_revenue.sql, we analyzed the monthly sales (item price excluding freight) and month-over-month growth from September 2016 to September 2018.

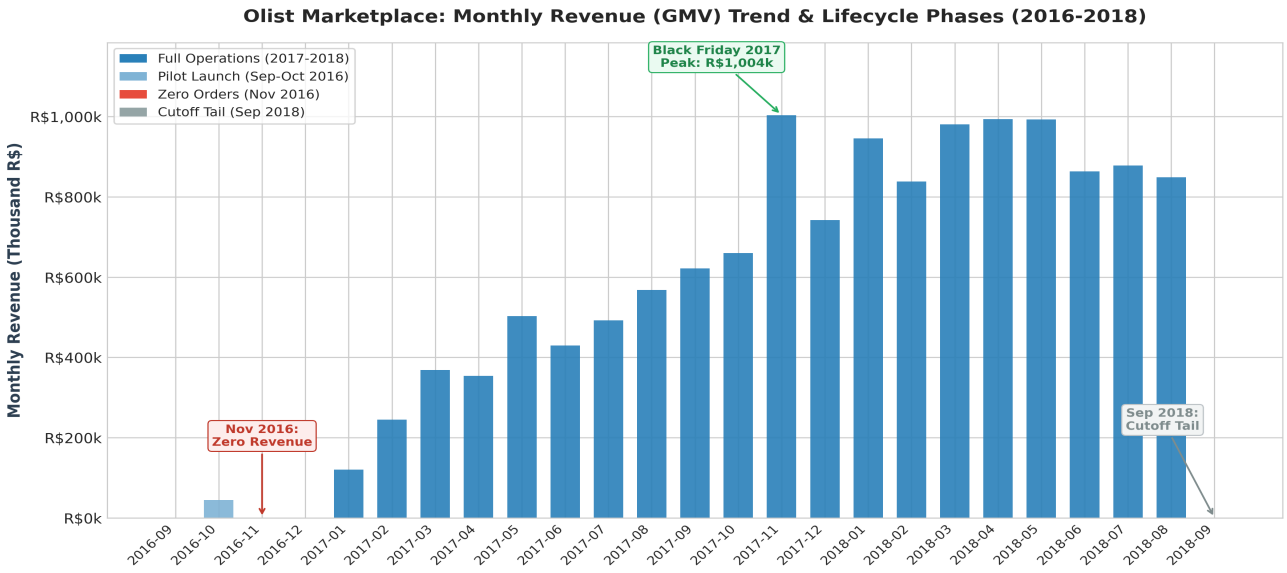


Figure 1: Monthly sales trend (GMV in R$) across the lifecycle phases of Olist.

Key Findings from Monthly Sales:

- **Pilot Phase (Late 2016):** The platform started with 295 test orders in Sep–Oct 2016 (R$ 44.7k total sales). November 2016 had exactly 0 orders because marketplace operations paused for backend upgrades. December 2016 had just 1 test order (R$ 10.90).
- **Hyper-Growth Phase (2017 to Early 2018):** Sales grew quickly from R$ 120.1k in January 2017 to over R$ 900k per month by early 2018.
- **Black Friday Spike (November 2017):** Sales jumped to R$ 977.7k (+42.2% month-over-month growth). On Black Friday, the marketplace handled a single-day record of 1,141 orders.
- **Peak and Plateau (Mid 2018):** The highest monthly sales happened in May 2018 at R$ 992.8k. After May 2018, sales leveled off between R$ 850k and R$ 880k per month.
- **Data Cutoff (September 2018):** September 2018 recorded only 1 order (R$ 145.00) because data collection ended at the start of that month.

5. Delivery Delays and Customer Reviews

Using SQL query 08_delivery_delay_vs_reviews.sql, we checked how delivery delays affect customer review scores (1 to 5 stars).

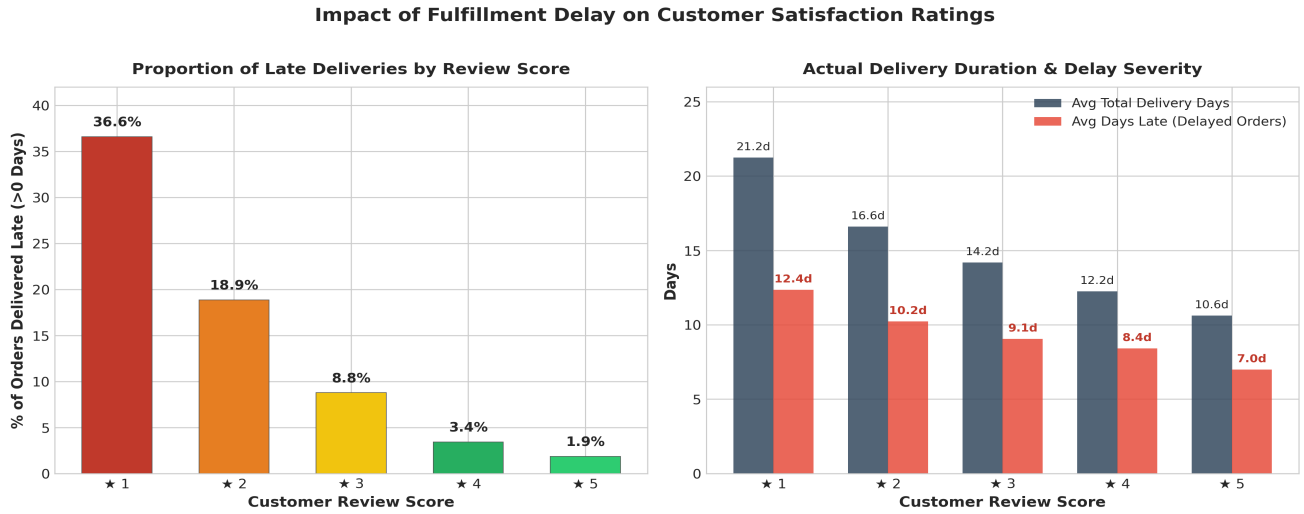


Figure 2: Percentage of late orders (left) and delivery duration vs delay severity (right) by review rating.

Review Rating	Total Reviews	Distinct Orders	% Late Orders	Avg Delivery Days	Avg Days Late (When Delayed)
1 Star	9,405	9,380	**36.60%**	21.25 days	**+12.36 days late**
2 Stars	2,941	2,938	18.86%	16.61 days	+10.23 days late
3 Stars	7,961	7,942	8.78%	14.20 days	+9.05 days late
4 Stars	18,987	18,943	3.44%	12.25 days	+8.41 days late
5 Stars	57,059	56,810	**1.86%**	10.63 days	+6.99 days late

Important Finding — Correlation vs. Causation:

- **Late delivery is a major cause of 1-star reviews:** 36.60% of 1-star reviews experienced late deliveries, and when late, the order was overdue by an average of 12.36 days (total delivery time took over 21 days). In contrast, only 1.86% of 5-star orders were late.
- **Delivery delay is NOT the only cause:** Notice that **63.40% of 1-star reviews arrived on time or early**. Customer review text shows that product defects, wrong items sent by sellers, and missing parts also cause many 1-star ratings. Therefore, fixing delivery delays is important, but checking seller product quality is also necessary.

6. Repeat Customers and Top Product Categories

This section covers findings from SQL queries 07_repeat_purchase_rate.sql, 06_rolling_7_day_orders.sql, and 05_top_categories_by_state.sql.

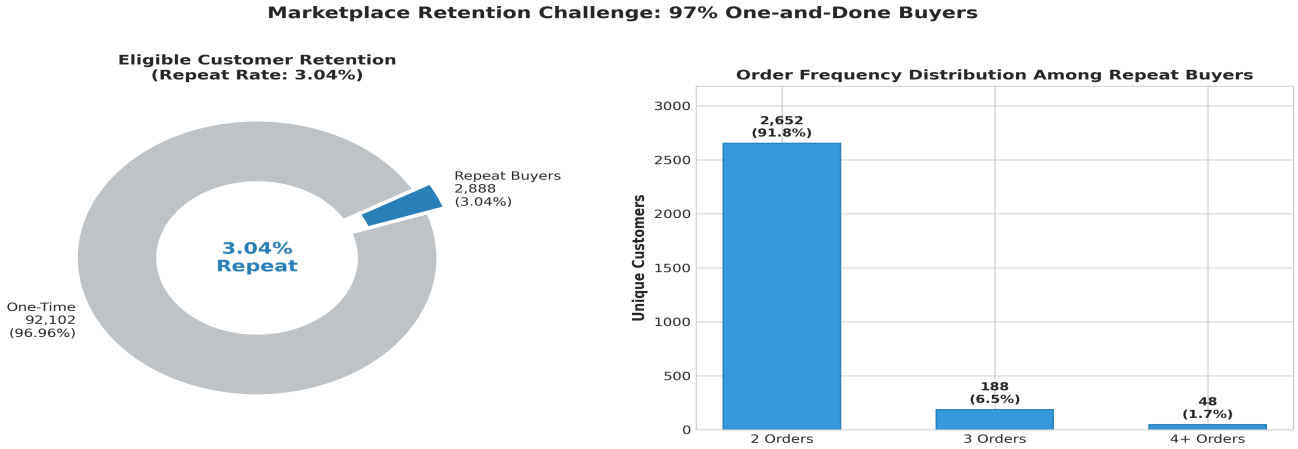


Figure 3: Repeat customer rate: 96.96% bought once vs. 3.04% who bought again.

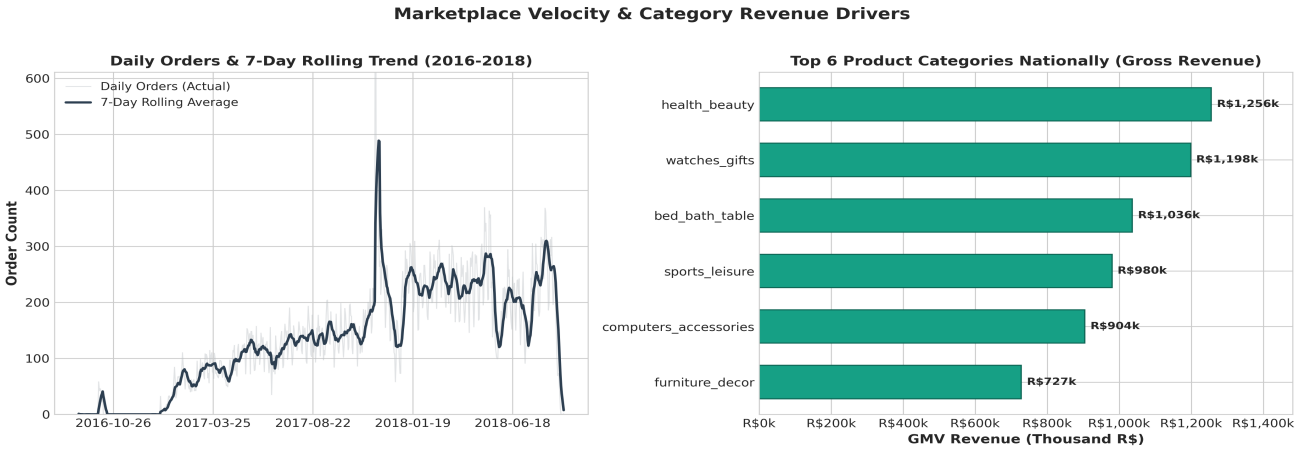


Figure 4: 7-day rolling order count (left) and top 6 grossing categories (right).

Key Findings:

- **Very Low Repeat Rate:** Out of 94,990 eligible buyers (using customer_unique_id), 92,102 customers (96.96%) placed only one order. Only 2,888 customers (3.04%) ever returned for a second order. Among repeat buyers, 91.8% placed exactly 2 orders, and very few placed 3 or more.
- **Daily Orders and Growth:** Daily order volume grew from under 20 orders per day in early 2017 to about 150–200 orders per day in 2018. The 7-calendar-day rolling average smoothed out weekend dips and clearly captured the Black Friday spike of 1,141 orders.
- **Top States and Categories:** São Paulo (SP) is the biggest state with over 35% of total sales, followed by Rio de Janeiro (RJ) and Minas Gerais (MG). The highest-earning categories across Brazil are *health_beauty* (R$ 1.26M), *watches_gifts* (R$ 1.21M), and *bed_bath_table* (R$ 1.04M).

7. Suggestions to Improve the Marketplace

Based on our analysis of delivery delays and repeat purchases, here are three practical suggestions:

1. Give More Realistic Delivery Dates:
Instead of showing fixed delivery estimates, add a few buffer days based on the distance between seller and buyer. If customers expect the order to arrive a little later, they will not be unhappy when small shipping delays occur. This can prevent many 1-star reviews caused by minor 1–2 day delays.

2. Enforce 48-Hour Seller Shipping:
Require sellers to pack and hand over items to the courier within 48 hours. Sellers who regularly take too long should be given lower visibility in search results. Also, using local shipping hubs in high-volume states like São Paulo and Rio de Janeiro will cut delivery transit time.

3. Send 14-Day Follow-Up Discounts:
Since 97% of buyers currently churn after one order, set up automated emails 14 days after delivery offering a 10% discount on their next order. Focus on replenishable products like Health & Beauty and Bed & Bath to encourage customers to make a second purchase.

8. Weekly Monitoring Metrics

Metric	Current Value	Target	Goal
Total Monthly Sales	R$ 848k/mo	**> R$ 1.0M/mo**	Maintain steady monthly sales growth.
Repeat Purchase Rate	3.04%	**> 6.00%**	Double the percentage of returning customers.
On-Time Delivery Rate	92.20%	**> 96.00%**	Reduce delivery delays across all states.
Average Days Late (Delayed)	+11.8 days	**< 3.0 days**	Bring down severe double-digit delivery delays.
Average Review Rating	4.09 / 5.0	**> 4.30 / 5.0**	Improve overall customer feedback.

9. Limitations

• **Seller vs Courier Time:** The dataset does not tell us how much time the seller took to pack vs how long the courier took to drive.
• **Missing Marketing Costs:** Ad spend and customer acquisition costs (CAC) are not included, so we cannot calculate acquisition ROI.
• **No Returns Data:** The dataset does not record product returns, exchanges, or warranty complaints after delivery.
• **No Website Traffic Logs:** We cannot see how many people visited the store or abandoned their shopping carts.